

Abstract Syntax Trees, Type Inference, and Operational Semantics

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1 Abstract syntax trees

An *abstract syntax tree* (AST) is a tree representation of the hierarchical structure of a program. Unlike a concrete parse tree, an AST omits purely syntactic detail such as parenthesis, semicolons used as separators, keywords whose meaning is already captured by the node type; retaining only the information needed to define the program's meaning.

Each *internal node* of the AST corresponds to a language construct (sequence, assignment, conditional, etc.) and each *leaf* corresponds to an atomic value or reference.

For an imperative language with mutable references, the relevant constructs are:

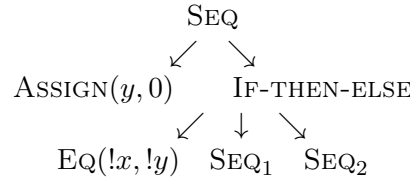
- $C_1 ; C_2$, sequential composition
- $l := e$, assignment to location l
- **if** B **then** C_1 **else** C_2 , conditional command
- **while** B **do** C , iteration
- $!l$, dereferencing a location (read the value stored at l)
- n, b , integer and boolean literals
- $e_1 \text{ op } e_2$, arithmetic or boolean combination

1.1 solved example

Build the AST of the following program:

$y := 0$; **if** $!x = !y$ **then** $x := !x + 1$; $y := !y - 1$ **else** $x := !x - 1$; $y := !y + 1$

The top level operator is the semicolon that separates the two main components, so the root is a SEQ node. Its left child is the assignment $y := 0$, and its right child is the conditional. The conditional has three children: the guard $!x = !y$, the *then*-branch $x := !x + 1$; $y := !y - 1$, and the *else*-branch (symmetric).



note that $!x$ is represented as a Deref node applied to the location x , which is distinct from x as an identifier: the location names a cell in the store, and dereferencing reads its content.

1.2 Exercises

- Build the AST of: $x := !x \times 2$; **if** $!x > 10$ **then skip** **else** $x := !x + 1$
- Build the AST of: **while** $!i < !n$ **do** $s := !s + !i$; $i := !i + 1$
- Reconstruct the source program corresponding to the following AST:
root is SEQ with left child ASSIGN($x, 5$) and right child IF-THEN-ELSE(EQ($!x, 5$), ASSIGN($y, 1$), ASS

2 Type inference

Type inference is the process by which a compiler deduces the type of an expression without explicit annotations from the programmer. the classical algorithm for the simply typed lambda calculus (and its extensions) is *Hindley Milner* (HM), which works by:

1. Assigning a fresh type variable α_i to each sub-expression whose type is unknown.
2. Generating *constraints* of the form $\tau_1 = \tau_2$ from the typing rules.

3. Solving the constraints by *unification*.

if unification fails, the program contains a type error; if it succeeds, the resulting substitution gives the most general type.

2.1 Typing rules for references

$\frac{\text{T-DEREF} \quad \Gamma \vdash !l : T}{\Gamma \vdash l : \text{ref } T}$	$\frac{\text{T-ASSIGN} \quad \Gamma \vdash l := e : \text{unit}}{\Gamma \vdash l : \text{ref } T \quad \Gamma \vdash e : T}$	$\frac{\text{T-WHILE} \quad \Gamma \vdash \text{while } B \text{ do } C : \text{unit}}{\Gamma \vdash B : \text{bool} \quad \Gamma \vdash C : \text{unit}}$
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2.2 solved example

$\lambda f. f f$ does not typecheck

Consider `proc(f) zero?((f f))`. We try to infer the type of f .

- Let $f : \alpha$.
- The expression $f f$ requires f to be a function type, so $\alpha = \alpha \rightarrow \beta$ for some fresh β .
- But $\alpha = \alpha \rightarrow \beta$ has no finite solution: expanding repeatedly yields $\alpha = (\alpha \rightarrow \beta) \rightarrow \beta = ((\alpha \rightarrow \beta) \rightarrow \beta) \rightarrow \beta = \dots$

unification fails (occurs check). Therefore the expression has no type in the simply typed lambda calculus without recursive types.

2.3 Scala generics

```
case class MyPair[A,B](x:A,y:B)
val p = MyPair(1, "Scala")
def id[1](x:T)=x
val q = id(1)
```

the definition of `id` introduces a **polymorphic** type: it can be applied to any type, and each call site independently instantiates a T .

```
B = !n < 5
B1 = !term <= 2
C1 = res := !n
C2 = c:=!a+!b; res:=!n; !c; a:=!b; b:=!c
C = if B1 then C1 else C2; n:=!n+1
a:=1; b:=1; while B do C
```

reading off types from the operations performed:

- $n, res, a, b, c : \mathbf{ref\ int}$ (assigned integers and dereferenced in arithmetic).
- $term : \mathbf{ref\ int}$ (compared with 2).
- $B, B_1 : \mathbf{bool}$.
- $C_1, C_2, C : \mathbf{unit}$ (commands produce no value).
- The whole program : **unit**.

2.4 Exercises

Infer the type of the S combinator: $\lambda x. \lambda y. \lambda z. x\ z\ (y\ z)$. What constraints does unification generate?

Consider $\lambda f. \lambda x. f\ (f\ x)$. Does it typecheck? If so, give its most general type.

What types are inferred for `f`, `result`, and `pairs` in the following Scala snippet?

```
def f[A](lst: List[A]) = lst.head
val result = f(List(1,2,3))
val pairs  = List((1,"a"), (2,"b"))
```

Given $x : \mathbf{ref\ int}$ and $b : \mathbf{ref\ bool}$, does the following expression typecheck? If so, give its type.

$b := !x = 0 ; \mathbf{if\ !b\ then\ } x := 1 \mathbf{\ else\ skip}$

3 Operational semantics

3.1 Big-step semantics

In *big-step* (natural semantics, a judgement) $\langle e, \sigma \rangle \Downarrow v$ asserts that expression e in store σ evaluates *directly* to value v , without exposing intermediate steps. Derivations are trees whose leaves are axioms and whose internal nodes are inference rules.

3.1.1 Lists: concat and append

We assume the constructors **nil** (empty list) and **cons**(h, t) (list with head h and tail t).

$\text{concat}(e, \ell)$, insert element e at the end of ℓ .

$$\begin{array}{c} \text{CONCAT-NIL} \\ \langle \text{concat}(e, \text{nil}), \sigma \rangle \Downarrow \text{cons}(e, \text{nil}) \end{array} \quad \frac{\text{CONCAT-CONS} \quad \langle \text{concat}(e, \text{cons}(h, t)), \sigma \rangle \Downarrow \text{cons}(h, t')}{\langle \text{concat}(e, t), \sigma \rangle \Downarrow t'}$$

$\text{append}(\ell_1, \ell_2)$ concatenate two lists.

$$\begin{array}{c} \text{APPEND-NIL} \\ \langle \text{append}(\text{nil}, \ell_2), \sigma \rangle \Downarrow \ell_2 \end{array} \quad \frac{\text{APPEND-CONS} \quad \langle \text{append}(\text{cons}(h, t_1), \ell_2), \sigma \rangle \Downarrow \text{cons}(h, \ell')}{\langle \text{append}(t_1, \ell_2), \sigma \rangle \Downarrow \ell'}$$

Derivation for append We verify $\langle \text{append}(\text{cons}(1, \text{nil}), \text{cons}(2, \text{nil})), \sigma \rangle \Downarrow \text{cons}(1, \text{cons}(2, \text{nil}))$:

$$\frac{\text{APPEND-CONS} \quad \langle \text{append}(\text{cons}(1, \text{nil}), \text{cons}(2, \text{nil})), \sigma \rangle \Downarrow \text{cons}(1, \text{cons}(2, \text{nil}))}{\text{APPEND-NIL} \quad \langle \text{append}(\text{nil}, \text{cons}(2, \text{nil})), \sigma \rangle \Downarrow \text{cons}(2, \text{nil})}$$

3.2 Small-step semantics

In *small-step* semantics, a judgement $\langle P, \sigma \rangle \rightarrow \langle P', \sigma' \rangle$ performs a single atomic reduction. A program is fully evaluated when no rule applies (it is a value or a stuck state) or when it reduces to **skip** (for commands).

3.2.1 For-loops in SIMP

The grammar of SIMP extended with **for** is:

$$C ::= l := E \mid C ; C \mid \text{if } B \text{ then } C \text{ else } C \mid \text{while } B \text{ do } C \mid \text{skip} \mid \text{for } (C; B; C) \{C\}$$

The cleanest approach is to treat **for** as *syntactic sugar*:

$$\text{for } (C_i; B; C_s) \{C_b\} \triangleq C_i; \text{while } B \text{ do } C_b; C_s$$

If the language requires explicit small-step rules (no sugar), we give:

$$\frac{\text{FOR-INIT} \quad \langle \text{for } (C_i; B; C_s) \{C_b\}, \sigma \rangle \rightarrow \langle \text{for } (C'_i; B; C_s) \{C_b\}, \sigma' \rangle}{\langle C_i, \sigma \rangle \rightarrow \langle C'_i, \sigma' \rangle}$$

FOR-SKIP

$\langle \text{for } (\text{skip}; B; C_s) \{C_b\}, \sigma \rangle \rightarrow \langle \text{if } B \text{ then } (C_b; C_s; \text{for } (\text{skip}; B; C_s) \{C_b\}) \text{ else skip}, \sigma \rangle$

Rule FOR-INIT reduces the initialiser one step at a time. Rule FOR-SKIP fires once the initialiser has been fully executed: it checks the guard and either runs the body followed by the step expression (and loops), or terminates with **skip**.

3.3 Exercises

- Write big-step rules for **reverse**(ℓ), which reverses a list.
- Use **concat** as a helper.

Prove with a full derivation tree:

$\langle \text{concat}(3, \text{cons}(1, \text{cons}(2, \text{nil}))), \sigma \rangle \Downarrow \text{cons}(1, \text{cons}(2, \text{cons}(3, \text{nil})))$

- Write big-step rules for **length**(ℓ), which returns the number of elements, and evaluate $\langle \text{length}(\text{cons}(1, \text{cons}(2, \text{nil}))), \sigma \rangle \Downarrow 2$ with a derivation tree.

4 Lambda calculus

5 Type safety

6 Denotational semantics

7 Axiomatic semantics (Hoare logic)